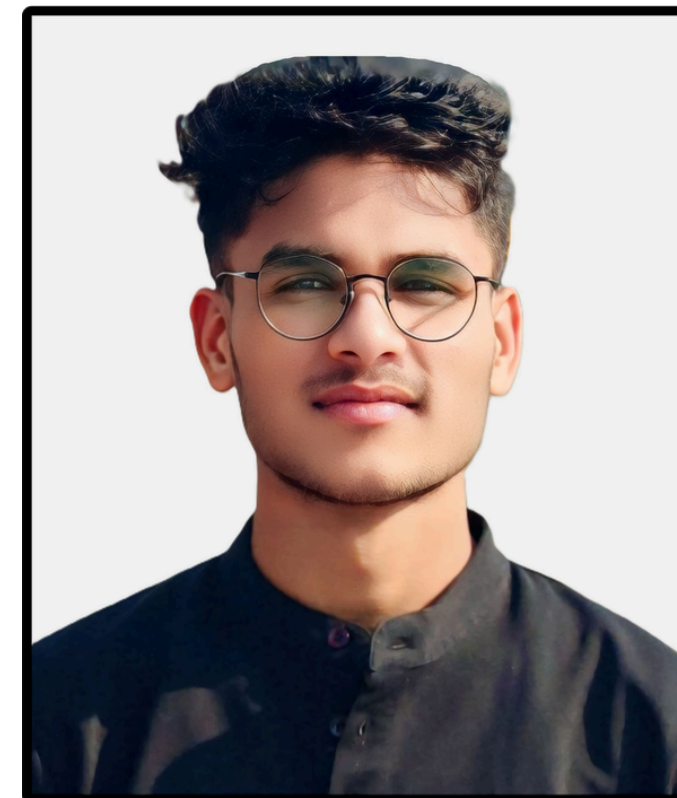


PROJECT TITLE: SUSTAINABLE TRAVEL PLANNER AGENT

Domain: Travel & Tourism / Sustainability
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SDG Alignment

Primary SDG:

- SDG 11: Sustainable Cities and Communities

Secondary SDGs:

- SDG 13: Climate Action
- SDG 12: Responsible Consumption and Production

Why this SDG?

Tourism contributes significantly to carbon emissions, resource consumption, and community displacement. This project promotes sustainable travel choices that benefit both the environment and local communities.

Problem Statement

How might we use AI to help travellers plan trips that minimise carbon footprint, support local communities, and promote sustainable transport and accommodation, so that tourism becomes more sustainable?

Current Challenges:

- Generic travel planners ignore environmental impact.
- Lack of real-time carbon footprint estimation.
- Over-tourism in popular destinations; neglect of lesser-known eco-spots.
- No easy way to find eco-certified stays and local businesses.
- Travellers lack awareness of sustainable alternatives.

AI Solution Overview

A Multi-Agent AI System using IBM Bob, IBM Granite, IBM watsonx.ai, and RAG.

Key Features:

- Suggests low-carbon destinations and off-beat locations.
- Builds itineraries using sustainable transport (trains, buses, shared rides).
- Recommends eco-certified stays and community-owned businesses.
- Estimates carbon footprint of each trip.
- Provides real-time alerts and sustainable travel tips.

Core Agents:

1. Destination Discovery Agent
2. Sustainable Itinerary Builder Agent
3. Green Transport & Stay Agent
4. Impact Tracker Agent

Target Users

- Eco-conscious travellers
- Students and families planning budget-friendly, sustainable trips
- Tourists visiting sensitive ecological regions
- Local tourism boards and sustainable businesses
- Travel platforms looking to integrate green options

Responsible AI Considerations

- Fairness: Promotes lesser-known destinations and local communities, avoiding over-tourism.
- Transparency: Clearly explains carbon estimates and sustainability scores.
- Ethics: Avoids greenwashing; only recommends genuinely eco-certified options.
- Privacy: Collects minimal personal data; no unnecessary sharing of location or preferences.
- Accountability: Human oversight for recommendations; clear disclaimers.

Expected Impact

- Reduction in travel-related carbon emissions.
- Increased support for local economies and community tourism.
- Greater awareness of sustainable travel choices.
- Scalable solution for cities, tourism boards, and travel platforms.
- Contribution to SDG 11, 12, and 13 targets.

Design Thinking – Empathize & Define

Empathize:

- Travellers want convenience but also care about sustainability.
- Local communities need fair tourism income.
- Environment suffers from unchecked tourism.

Define:

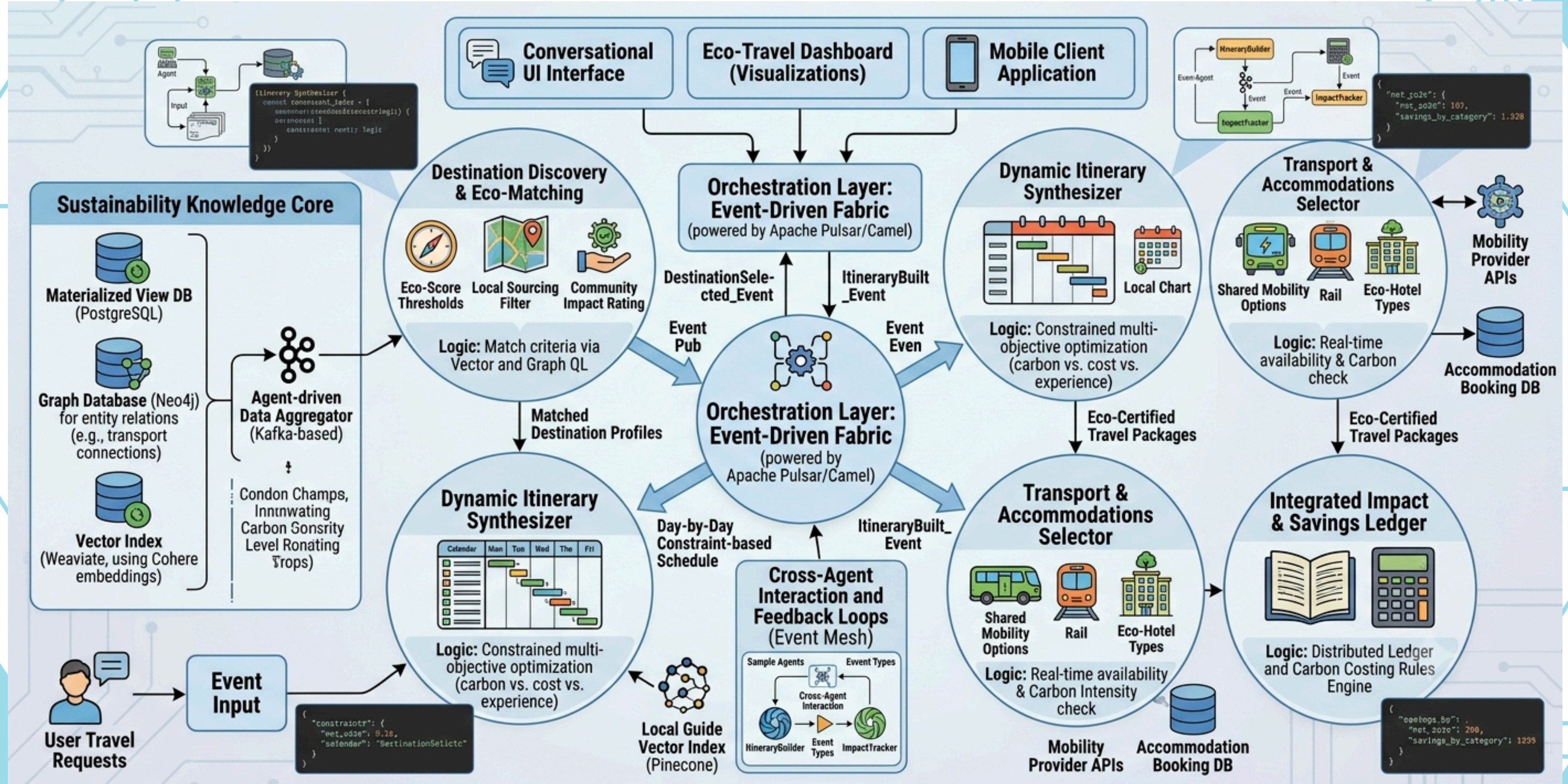
- Current travel planners lack integrated sustainability features, carbon tracking, and local community support.
- Need for a smart, AI-driven solution that balances user preferences with environmental and social impact.

Design Thinking – Ideate

Brainstormed AI Solutions:

- Multi-agent AI with RAG for real-time sustainable travel data.
- Carbon footprint calculator integrated into itinerary planning.
- Recommendation engine for eco-certified stays and local businesses.
- Conversational interface for easy user interaction.
- Dashboard for visualising carbon savings and community impact.

Design Thinking – Prototype



Design Thinking – Test & Refine

Testing Approach:

- Mentor feedback on agent logic and ethics.
- User testing for clarity, usability, and usefulness.
- Refinement of carbon estimation models and recommendations.
- Ensuring responsible AI guidelines are met.

Prototype / Demo – Prompt Workflow

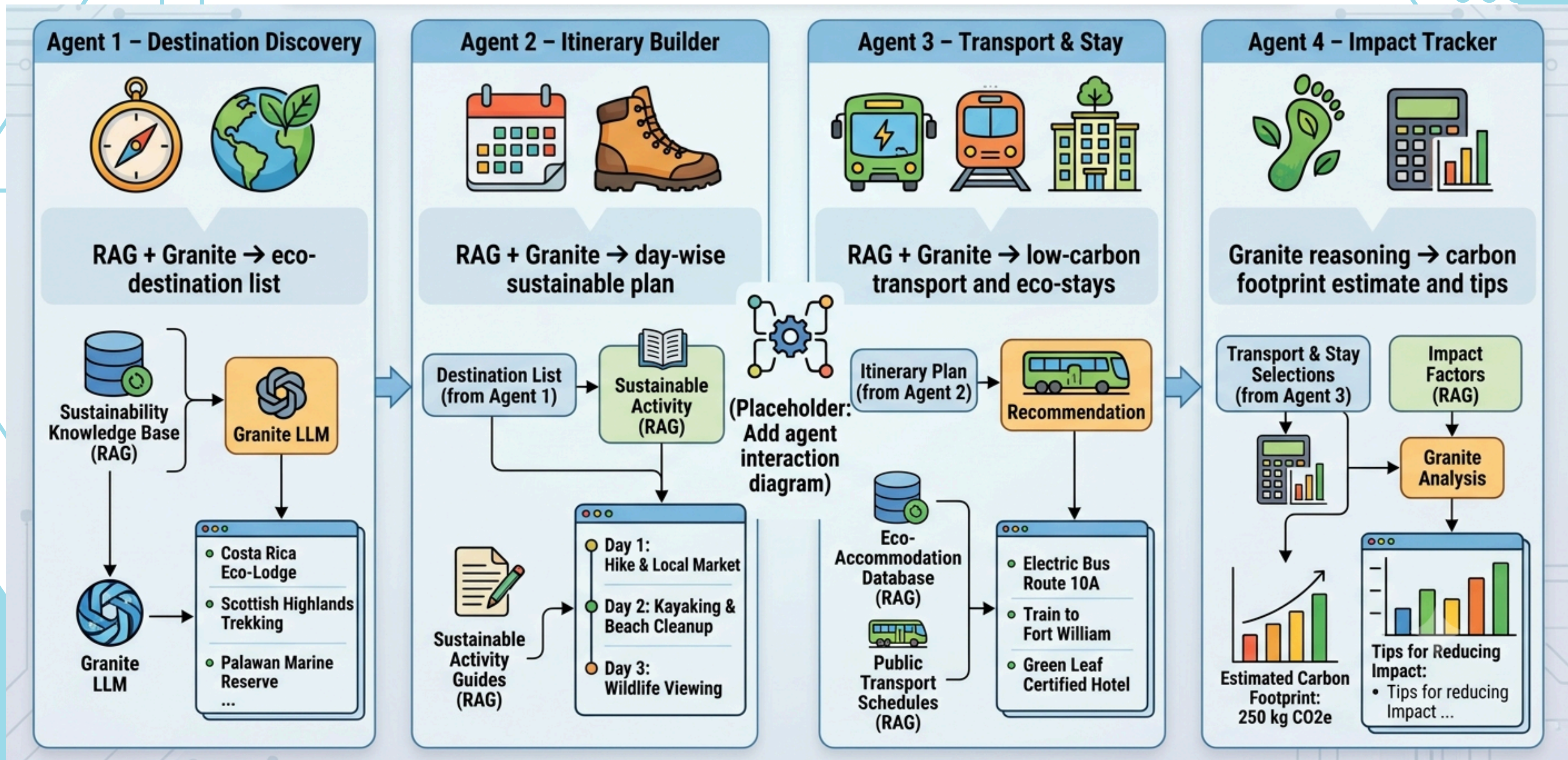
Example Prompt:

“Plan a 3-day trip from Delhi to a nearby hill station with minimal carbon footprint.”

Workflow:

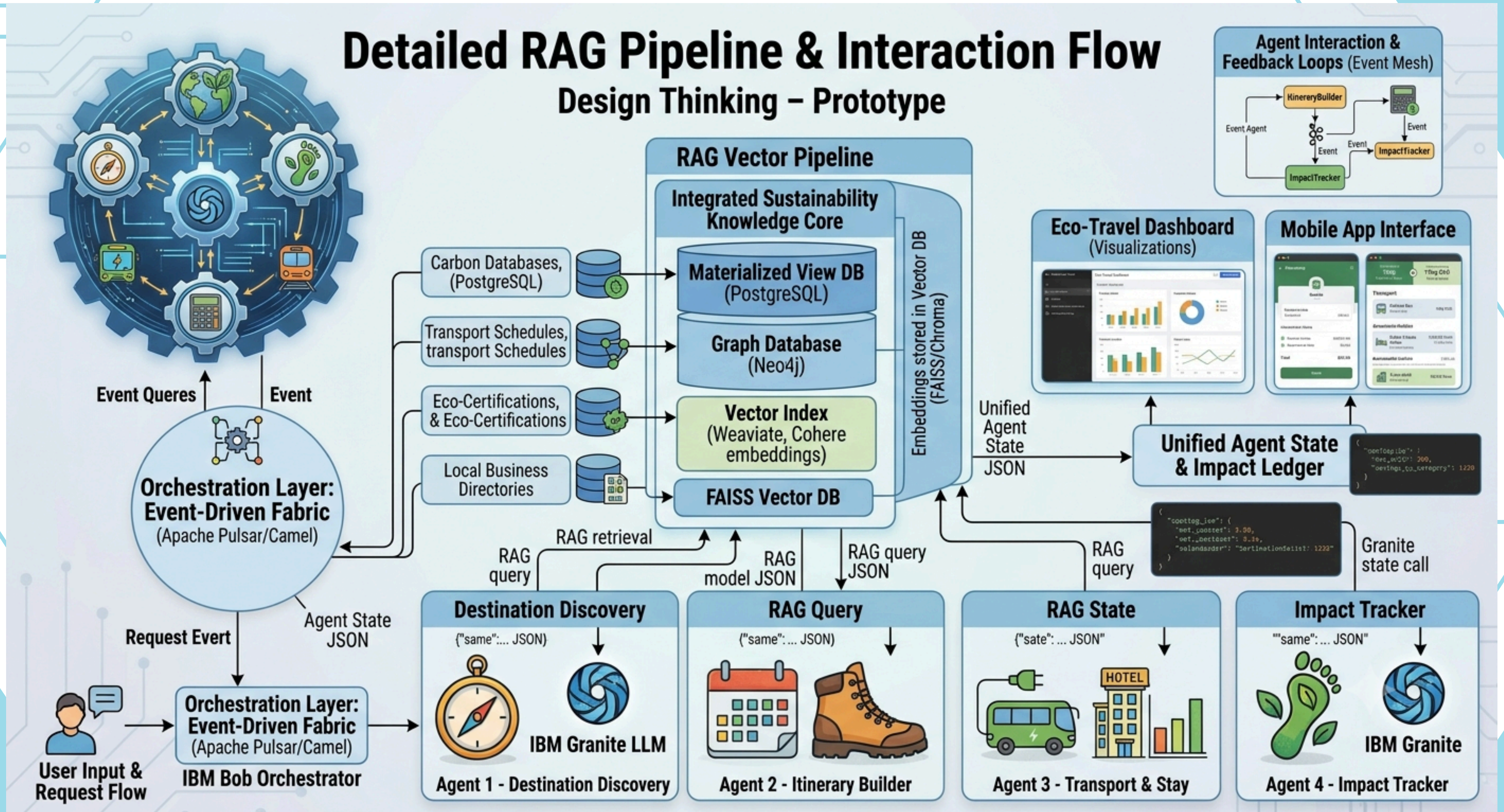
1. IBM Bob Orchestrator routes request.
2. Destination Discovery Agent retrieves eco-friendly hill stations via RAG.
3. Sustainable Itinerary Builder Agent creates day-wise plan using trains/buses, local walks, eco-activities.
4. Green Transport & Stay Agent recommends eco-homestays and local eateries.
5. Impact Tracker Agent estimates carbon footprint and suggests improvements.
6. IBM Granite generates natural language itinerary with explanations.
7. Dashboard displays map, carbon score, and local community benefits.

Prototype / Demo – Agent Logic

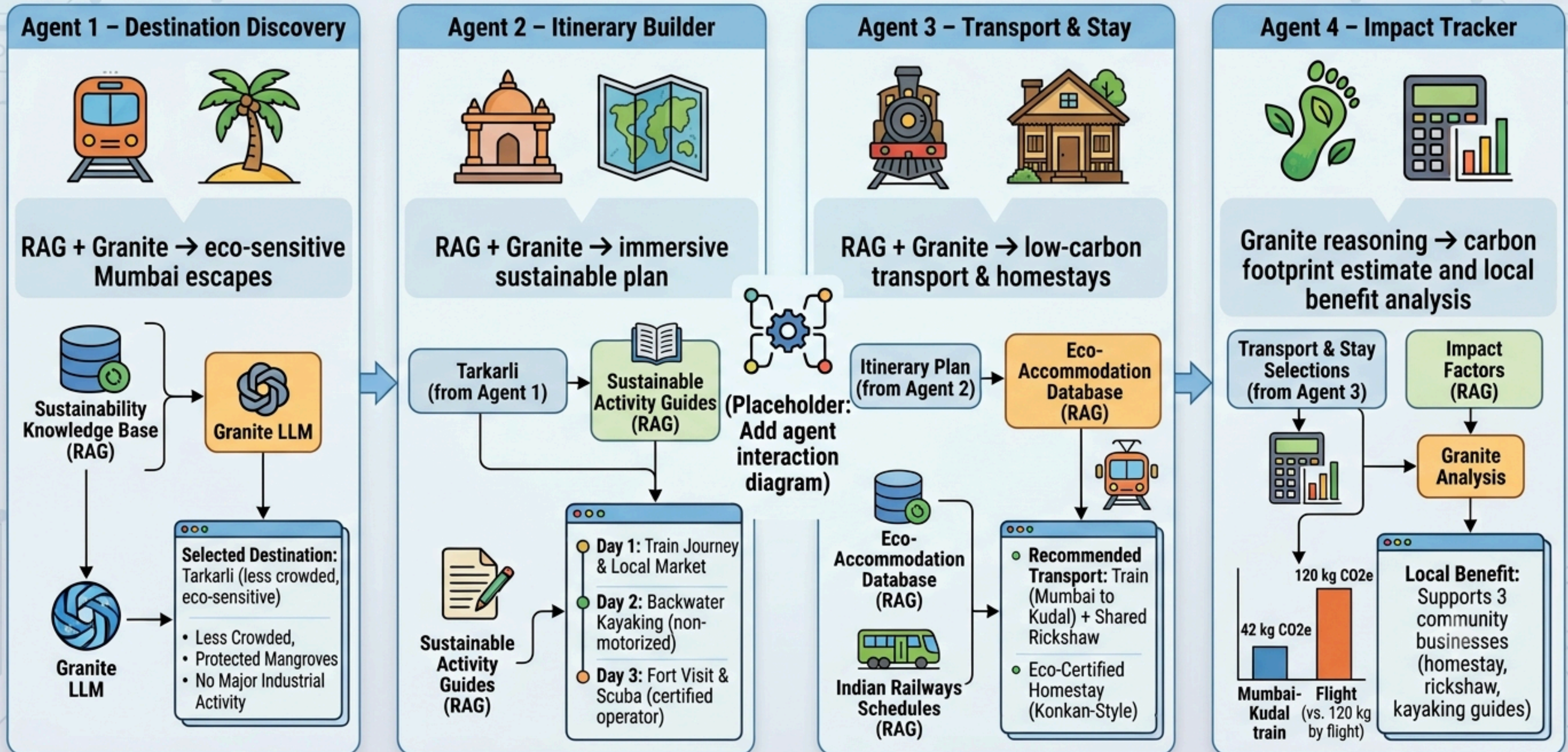


Detailed RAG Pipeline & Interaction Flow

Design Thinking – Prototype



Prototype / Demo – Sample Input / Output: Mumbai-Tarkarli Eco-Trip



Results

Input Query:.

```
Distance = 350 km (Delhi to Binsar)  
Transport Modes = Train vs Flight
```

Output:.

```
Carbon Footprint Estimate (train): 17.5 kg CO2  
Carbon Footprint Estimate (flight): 52.5 kg CO2  
Carbon Savings: 35.0 kg CO2
```

Results

Input Query:.

```
1 Plan a 3-day trip from Delhi to a nearby hill station with  
minimal carbon footprint.
```

Output:.

```
Sustainable Travel Plan for: Plan a 3-day trip from Delhi to a nearby hill station with minimal carbon  
footprint.
```

```
Destination: Binsar (Eco Rating: 4.8/5)
```

```
Transport: Train from Delhi to nearest station, then shared taxi to Binsar.
```

```
Stay: Eco-certified homestay in Binsar (community-owned).
```

```
Itinerary:
```

- Day 1: Arrive in Binsar, check into eco-homestay, local walk.
- Day 2: Nature trail, visit local village, organic farm lunch.
- Day 3: Sunrise point, local craft workshop, depart.

```
Carbon Footprint Estimate (train): 17.5 kg CO2
```

```
Carbon Footprint Estimate (flight): 52.5 kg CO2
```

```
Carbon Savings: 35.0 kg CO2
```

```
Local Benefit: Supports 3 community businesses.
```

Results

Input Query:.

Plan a 3-day trip from Delhi to a nearby hill station with minimal carbon footprint.

Output:.

Destination: Binsar (Eco Rating: 4.8/5)

Transport: Train + shared taxi

Stay: Eco-certified homestay (community-owned)

Carbon Savings: 35.0 kg CO2

Local Benefit: Supports 3 community businesses

Itinerary:

Day 1: Arrive, eco-homestay, local walk

Day 2: Nature trail, village visit, organic farm lunch

Day 3: Sunrise point, craft workshop, depart

Impact Statement

If implemented:

- Travellers make informed, low-carbon choices.
- Local communities and sustainable businesses gain visibility and income.
- Tourism pressure shifts from over-crowded destinations to lesser-known areas.
- Overall carbon emissions from travel are reduced.
- Awareness of responsible tourism increases.

Who benefits?

- Environment: lower emissions, less ecological damage.
- Travellers: personalised, budget-friendly, sustainable trips.
- Local communities: economic empowerment and cultural preservation.
- Governments & tourism boards: data-driven insights for sustainable tourism planning.

Future Enhancements

- Integration with wearables and travel devices for real-time data.
- AI-powered voice assistant and multilingual support.
- Dynamic replanning due to weather or disruptions.
- Group travel planning for families and corporate groups.
- AR/VR destination previews for immersive experiences.

FLOW PROCESS: PROBLEM IDENTIFICATION TO RESPONSIBLE AI & ETHICS



1. Problem Identification

Pinpointing real-world issues, gathering data, and defining scope

2. SDG Alignment

Mapping the problem to relevant UN SDGs and global targets

3. AI Role & Ideation

Brainstorming innovative AI solutions, models, and technical approaches

4. Design Thinking

Empathizing with users, defining needs, and ideating user-centric solutions

5. Prototyping & Testing

Building iterative models, creating MVPs, and gathering feedback

6. Impact & Evaluation

Measuring social, environmental, and economic impact; assessing performance

7. Responsible AI & Ethics

Ensuring fairness, accountability, transparency, privacy, and mitigating bias

THANK YOU!

